

6. MASS AND BALANCE

6.1 EQUIPMENT LIST ENTRIES

System/Equipment Designation	Part No.	Mass (kg)	Arm (mm)	Massmoment (kgmm)
Heating and Ventilation System	105-80052	13.6	1400	19040
Heating and Ventilation System	D134-642	13.6	1400	19040
Heating and Ventilation System	105-80050	(13.7)	(1200)	(16440)
Ventilation System	105-80053	10.0	1150	11500
Electric Heating Kit	105-80063	8.0	1300	10400
Electric Heating Kit	105-80067	(8.0)	(1100)	(8800)
Cargo Compt Heating Kit	105-80046	13.5	2100 (1850)	28350 (24980)
Cargo Compt Heating Kit	D134-641	13.5	2100 (1850)	28350 (24980)
Cargo Compt Heating Kit	D136-110	13.5	2100 (1850)	28350 (24980)
NOTE Above values in parentheses apply to Model S Variants.				

7. SYSTEM DESCRIPTION

The heating and ventilation system comprises an air source selector valve, a blower fan, four electric heater units, associated ducting, and system controls and indicators (see Fig. 1).

NOTE The ventilation system, P/N 105-80053, is comprised of the same components as the heating and ventilation system minus the four electric heater units.

A lever on the floor to the forward right of the copilot seat is provided for selecting the source of air for circulation; either fresh ram air or recirculated cabin air. The cabin air position (lever full forward) should be used to acquire maximum heating capability.

The four 1-kW electric heater units are normally located in the cockpit lower windshield area; two LH and two RH inner and outer units.

A control switch, a four-segment indicator light and a temperature selector knob are on the overhead panel. The three-position toggle switch (BLOWER-OFF-HEATING) is used to activate the system. With the switch in the HEATING position, 28V DC is supplied from the Auxiliary Bus via a circuit breaker (HEATING) on the switch panel to a temperature controller. The temperature controller (dependent upon the temperature selector knob

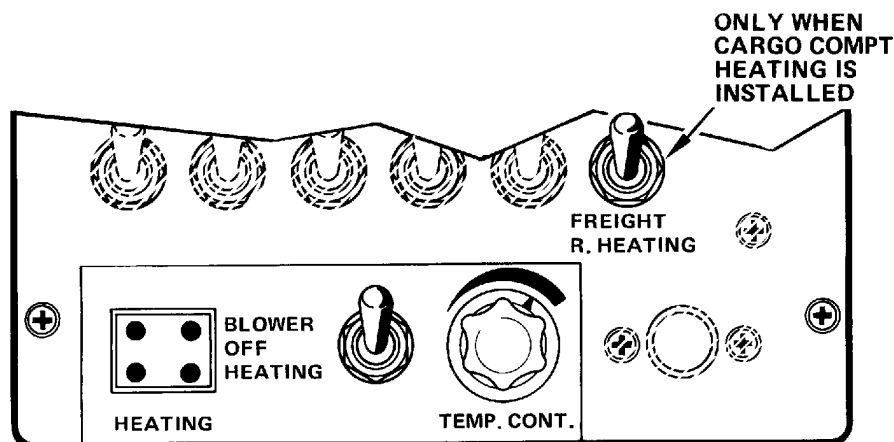
position) in turn supplies 28V DC from the Starter Bus via a fuse in the Main Relay Box to power the four separate heater units, their respective indicator light segments, and the blower.

NOTE Heater operation is possible only when both generators or an external power unit are supplying electric power due to the heater control power being supplied from the Auxiliary Bus.

The cabin may be ventilated with fresh air provided from a ram air inlet on the lower fuselage nose shell. Placing the select lever full aft supplies fresh air to the air distribution ducting. Placing the three-position toggle switch in the BLOWER position will supply 28V DC from the Main Bus via a circuit breaker (BLOWER) on the switch panel to power the blower fan.

Installation of the Electric Heating Conversion Kit integrates the four 1-kW heater units and the heating controls and indicators with the basic cabin ventilation system.

Installation of the Cargo Compartment Heater Conversion Kit removes the outer LH heater unit and a separate combined 1-kW heater/blower unit is installed on the RH cargo compartment sidewall. A separate toggle switch on the overhead panel is used to activate the cargo compartment heater after the cabin heating is switched on. The cargo compartment heater is not affected by the temperature controller, however, has its own overtemperature protection circuit. Activation of the overtemp switch is indicated by the respective indicator light segment on the overhead panel going off.



TYPICAL OVERHEAD PANEL

Fig. 1 Controls and Indicators